

Research Papers

Geographic Information System-based Multi-Criteria Decision-Making analysis for assessing prospective locations of Pumped Hydro Energy Storage plants in Morocco: Towards efficient management of variable renewables

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ABSTRACT

The considerable potential offered by wind and Solar Photovoltaic (SPV) energy, at competitive costs, constitutes a real opportunity to reduce CO₂ emissions, thus contributing to significant decarbonization. Nevertheless, these sources require energy storage, which remains a key solution to mitigate their intermittency and variability, as they are dispatchable energy production systems. Among these latter, Pumped Hydro Energy Storage (PHES) stands as the most widely adopted technology for utility-scale applications, since it offers numerous advantages. Finding a suitable site for a PHES plant, however, remains a challenge for new project development, as it is highly dependent on topographic and geological site conditions. In this perspective, this paper presents an approach aiming at identifying and classifying appropriate locations for PHES plants based on the Analytic Hierarchy Process (AHP) method along with Geographic Information System (GIS) tools - ArcGIS 10.8. Considering the various PHES topologies, this assessment seeks to explore two: the first uses existing dams and lakes as primary reservoirs (Topology - T2), while the second considers the sea as a lower reservoir (Topology - T4). To our knowledge, this study introduces for the first time two novel decision criteria: evapotranspiration and sea surface salinity. These parameters lead respectively to evaporation losses from the storage reservoirs and corrosion effects on the different components of the installation. The proposed methodology was then applied to the case of Morocco, whereby the results revealed that a total of 1606 potential sites were identified, from which >36 % were deemed highly suitable. Furthermore, a total energy storage capacity of 6862 GWh was estimated from those locations, where about 72 % is represented by T2. This research work, the first of its kind for Morocco, aimed at securing cost-effective sites to help policy- and decision-makers develop strategies tailored to the country's conditions for a wide integration of variable renewables.

1. Introduction

Global energy demand and CO₂ emissions decreased by 4 % and 5.8 % respectively in 2020 due to Covid-19, representing the highest absolute decrease ever registered. In contrast, renewable energies were the

only energy source for which demand increased by 3 % in 2020 regardless of the pandemic. According to the IEA report [1], in 2021, energy demand recovered by 4.6 %, making global demand 0.5 % above the pre-Covid-19 level. In this perspective, renewable energies such as wind and Solar Photovoltaic (SPV) constitute the backbone of a

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successful energy transition. This latter, which is still ongoing, contributes to a significant decarbonization of the various sectors, i.e. industry, transport and buildings. These resources provide substantial global potential at competitive costs with reference to all fossil fuel generation sources [2]. By the end of 2020, the global SPV and wind capacity additions reached 135 GW and 114 GW, respectively, representing a 23 % and 90 % year-on-year expansion [3]. However, the growing share of these energies will have a significant impact on the grid operation, given their intermittent profile as they produce energy when the sun is shining or the wind is blowing. Energy storage is therefore required to support the large penetration of these energies as well as to ensure an adjustment between supply and demand. Among the various existing storage technologies are: Compressed Air Energy Storage (CAES), Flywheels, Batteries, Thermal Energy Storage (TES), Hydrogen, and Pumped Hydro Energy Storage (PHES) [4]. The latter represents the most mature and oldest technology, while also being increasingly used for large-scale energy storage. PHES systems typically consist of two reservoirs situated at different elevations. Water is pumped from the lower reservoir to the upper one, consuming electricity when demand and electricity prices are low. During peak periods, water drains to the lower reservoir through turbines, producing electricity when demand increases and electricity prices are high (Fig. 1). The PHES technology provides the advantage of operating on an instantaneous scale from minutes and hours to seasonal scale for long term storage [5]. Regardless of this, it bears noting that the technology has high energy efficiency, fast start-up, adjustable power output according to demand, as well as an extremely long technical lifetime [6]. At present, PHES leads the market in terms of total installed capacity, with a 96 % share, representing 176 GW global capacity as of mid-2017. The remaining share is occupied by thermal storage, batteries and mechanical storage (CAES and flywheels) [7]. Given the increasing share of intermittent renewables, PHES systems may serve to increase flexibility, absorb surplus generation while minimizing losses, balance variability, and thus ensure the stability of the power grid [5]. Installing a wind or SPV plant alongside the PHES system offers the potential to store energy on site, ensuring a reliable supply (Fig. 1). Another scheme that is gaining interest is a floating SPV system placed in the upper and lower reservoirs of a PHES plant,

creating a hybrid design that can leverage existing transmission infrastructure, enhance solar cell efficiency through water cooling, and minimize evaporative losses by shading the water (Fig. 1) [5]. Over and above these benefits, numerous types of layouts (Topologies - Ti) do exist, classifying PHES systems by nature of the reservoirs or water bodies. These different schemes are summarized by the Strategic Energy Technology Information System (SETIS) expert group into seven widely used topologies, as described in Table 1. As a matter of fact, numerous researchers around the world have been interested in identifying suitable sites for PHES development by exploring its different topologies. Several researchers among them have identified potential sites based on freshwater resources of slightly or moderately saline water, notably the following: J.A. Jimenez Capilla et al. [8] applied Geographic Information System (GIS) methods and Analytic Hierarchy Process (AHP) multi-criteria analysis to identify the optimal location of the upper reservoir to form a PHES system in the southern region of Spain while taking into

Table 1

Presentation of the various PHES topologies [24,25].

Topologies (Ti)	Approach description
T1	Connecting two existing reservoirs by one or more penstocks and associating a power plant
T2	Turning an existing lake or reservoir into a PHES by detecting a suitable site for a second reservoir. The second reservoir might be located on a flat or non-sloping area (i.e. by excavating or constructing shallow dams), on a depression or in a valley.
T3	A greenfield PHES based on an appropriate topographic setting: whether valleys that can be dammed, depressions, hilltops that could be slashed, etc. Such a topology is broader, i.e., it is not based on existing lakes or reservoirs and does not assume a flat area for construction of the second reservoir.
T4	Utilizing the sea as a lower reservoir with a new reservoir nearby.
T5	Multi-reservoir systems comprising both PHES and conventional hydropower.
T6	Large river is essentially the lower reservoir, which provides sufficient inflow to the PHES system.
T7	Using an abandoned mining pit as a base for the PHES. Its methodology would be similar to that of T2.

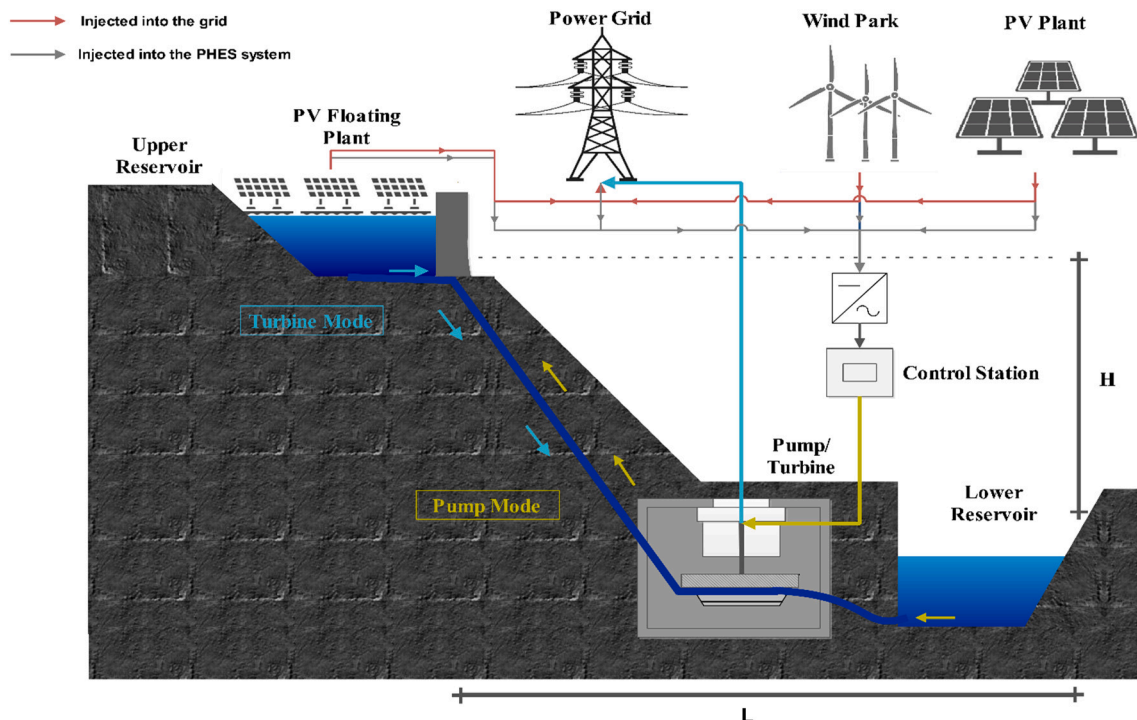


Fig. 1. PHES power plant operating principle.

account a number of construction and economic criteria that would make its construction feasible. The outcomes highlight approximately 470 ha of ideal sites in the study area. B. Lu et al. [9] have carried out a study for South Australia that seeks to locate potential sites for off-river PHES for two typical site typologies, dry-gully and turkey's nest, through a series of advanced GIS algorithms. The generated results identified 168 dry gully and 22 turkey nest sites with a total water storage capacity of 276 GWh. U. Nzotcha et al. [10] have selected the best sites among several PHES plants pre-selected for deployment of such a system in Western Cameroon, using Multi-Criteria Decision Making (MCDM) methods: AHP, fuzzy logic based scoring systems, and ELECTRE II (ELimination and Choice Expressing REality) involving sixteen criteria. Five sites, prioritized for power generation policy, were retained: (1) Enep, (2) Mbatu-dam, (3) Benakuma, (4) Bamendjin, (5) Mentchum. M. Stocks et al. [11] evaluated the global potential of PHES systems using GIS and high resolution global digital elevation models (DEM) data. This assessment identified a global potential comprising 616,000 sites with a storage capacity of 23,000 TWh.

Prospecting sites for small-scale PHES plants have likewise been considered in view of their lower impact on the environment. Among others, T. Soha et al. [12] assessed the possibilities of small-scale PHES sites within less favorable hilly and mid-mountain regions, and former mining areas, using GIS approach in mid-mountain areas in Europe. The findings revealed that at least 1590 MWh of electricity could theoretically be stored in the total of 15 upper reservoirs. Similarly, A. Rogeau et al. [13] have estimated the potential of small-scale PHES, through a designed generic method, taking into account two topologies: the connection between existing lakes and natural land depressions. Based on this method, a total potential comprised between 16 and 54 GWh has been estimated for the case of France, according to the adopted restrictions.

A further topology, T4, that received particular interest, was using seawater as the lower reservoir, as outlined in Table 1. The earliest application of such plant is the Okinawa Yanbaru PHES plant in Japan, built in 1999 [14]. This has led to many studies identifying appropriate sites for this kind of topology worldwide. N. Ghorbani et al. [15] assessed the technical potential of PHES system in Iran from four topologies through automated models based on GIS and TOPSIS method for the discovered feasible sites. The results reveal that the country has a technical potential from the studied topologies that reaches 5108 GWh. Y. Wu et al. [16] identified the best sites for seawater PHES in China, using eighteen evaluation criteria of different aspects, based on the VIKOR multi-criteria model. Fu Ying Island in Ning De was the best location for the construction of such a system. A. Pradhan et al. [17] detected prospective locations for seawater-PHES sites on the island of Curacao, using a developed GIS application. Two types of sites were explored, dry-gully and turkey's nest where its most interesting sites have a total energy storage capacity of 158.31 and 154.09 MWh respectively. Further studies have combined this topology with SPV and/or wind power by identifying appropriate sites for such a hybrid system. U. Portero et al. [18] presented a hybrid seawater-PHES-wind system on the coast of the island of Gran Canaria, taking advantage of seawater instead of using freshwater reserves. Thanks to the integration of the seawater unit, the wind-hydro system meets 29 % of the electricity demand compared to a wind system. Y. Wu, et al. [19] have identified the optimal site for wind-PV-seawater PHES plants in China considering the main factors that affect the system from sustainable development point of view, according to the fuzzy TODIM method. As a result, Qingtian Bay in Zhou Shan is selected as the best suited site to implement a wind-PV-seawater PHES integrated power plant. S.H.R. Ahmadi et al. [20] combined GIS, Analytic Network Process (ANP) and Fuzzy VIKOR to identify suitable sites for the construction of a wind-PHES plant in Kermanshah province, western Iran. Various technical, financial and environmental criteria are considered for this purpose. The findings revealed that the GilaneGharb dam is the most promising, within the studied area, for the development of the wind-PHES system,

providing a power capacity of about 31 MW.

Being the primary form of providing electricity storage services to the power grid and the largest source of installed storage capacity, the PHES system, despite the emerging storage technologies that are gaining interest around the world, remains the most widely adopted one. Nevertheless, there are some drawbacks that limit the future development of these systems, notably the challenge associated with finding suitable locations given their high dependence on site-specific geological and topographical conditions. Among others, these include the water availability, the significant difference in elevation between the upper and lower reservoirs, and the land suitability to build either the upper or lower reservoir to form the PHES plant [21]. These and other limitations hinder the expansion of such systems to a large enough scale to meet future bulk storage demands of power system operations across the globe.

In light of this concern, this paper intends to expand the existing literature on this topic with a comprehensive assessment aimed at identifying appropriate locations for the development of PHES plants. This is being carried out through an integrated approach combining an MCDM method with GIS tools following two topologies: T2 and T4, at the scale of a country - the case of Morocco. With this methodology, the different existing water bodies within the country (dams and lakes) are exploited to form a PHES plant according to T2 (Conventional-PHES). While the country's coastline, which extends over 3500 km, has been used to develop a PHES plant under T4 (Seawater-PHES) by exploiting the sea as a lower reservoir. For this purpose, the physical - realizable - potential was identified at first, while taking into account the specifications and restrictions of the model. Then, the various criteria affecting the development of these systems, from a technical, economic and environmental point of view, were considered to evaluate and classify the candidate locations by the so-called AHP method. The latter technique enables to seek the best results in complex configurations with diverse criteria. Indeed, the MCDM methods afford to stakeholders the opportunity to make the ideal decision by considering all factors at once. This therefore makes them widely applicable in the field of energy planning [22]. The resulting weights from the AHP analysis were subsequently combined with GIS-based modeling tools to generate the ranking maps and accordingly provide their respective spatial visualization. These techniques have been demonstrated to be very effective in evaluating the various renewable energy potentials [23]. Throughout this investigation, the ArcGIS 10.8 software was adopted with its various tools and features.

The novelty of this investigation lies in the incorporation of two factors in the evaluation of the selected sites, namely evapotranspiration and sea surface salinity. These latter have not been used in previous similar studies to the best of our knowledge, respectively for T2 and T4. These decision-making criteria were introduced in order to: (1) give priority to sites with lower evapotranspiration, considering the losses incurred for the storage reservoirs; (2) promote coastal regions with lower salinity, in order to reduce the corrosion of the different components of the seawater-PHES plant. Notwithstanding the fact that this is the first assessment work regarding the potential of PHES systems in its different topologies within the Moroccan case.

In light of the ever-increasing interest in variable renewables, the topographical and geological diversities of the investigated area (Morocco), together with the limited detailed studies on the topic, the authors were motivated to undertake the present study with the following main objectives:

- Develop a comprehensive methodology that may be extended to any area by only adjusting the related specifications and restrictions.
- Expand the decision criteria with additional parameters such as those mentioned earlier, so as to guarantee more accurate results and, therefore, cost-effective sites.
- Provide a representative overview of the most promising sites for the development of additional PHES systems in Morocco, aiming to

support policy and decision makers in implementing energy planning strategies adapted to the country's conditions.

This document is structured into three main sections as follows: (1) the first one presents the developed methodology comprising the specifications and restrictions of the model, the AHP decision making method as well as a depiction of the different criteria involved. (2) The second section introduces the application case study and the process of data collection and analysis. (3) Last section is entirely devoted to the presentation and discussion of the achieved results regarding the potential of PHES plants under both T2 and T4.

2. Methodology

In this paper, an approach is considered to identify candidate areas for the siting of PHES plants, according to two topologies: T2 and T4, taking into account existing water bodies. These locations undergo several topographic, occupation and environmental restrictions before being classified according to a set of technical, economic and environmental criteria. These were established in accordance with the specialists' perspectives and the studies already published in the literature. The subsequent parameters were analyzed using a decision-making method before employing GIS tools for collecting, processing and visualizing the potential sites. This method involves the following main steps:

1. Collection and preparation of the water bodies database (existing reservoirs, dams and lakes, ocean and sea).
2. Assessment of the physical potential -realizable- by applying the model requirements along with the restriction areas within a 16 km graphic buffer zone.
3. Establishment of criteria related to the two topologies under multiple aspects: technical, economic and environmental.
4. Application of the AHP multi-criteria decision-making method to the predefined criteria through a pairwise comparison, drawing on expert judgment and state-of-the-art literature review.
5. Identifying the prospective areas with the greatest accumulated weight according to the adopted method, through the application of the various GIS tools.

For the sake of illustration, ArcGIS 10.8 was employed to collect, analyze and process the entire database being used. The tools involved in this study are present in ArcToolboxes and in the ModelBuilder feature. In fact, the methodology proposed in the present study is applicable to any given area of interest by adapting the specifications associated with it.

2.1. Model specifications and restrictions areas

In order to conduct the multi-criteria evaluation process, it is first necessary to ensure that the potential site meets the topographic requirements for constructing a PHES plant. For that purpose, and since the T2 layout uses existing lakes and dams, a limit of 1,000,000 m³ was deemed necessary to ensure a reliable evaluation. However, this limit is ignored for seawater-based PHES systems, since seawater is abundant. Subsequently, and for either topology, a maximum search radius of 16 km was undertaken for each water body to locate candidate sites. This radius corresponds to areas that do not exceed the gross head and/or length-to-height ratio thresholds considered in the study (Table 2). Within this 16 km buffer zone, sites meeting a gross head ranging from a minimum of 100 m to a maximum of 800 m were considered appropriate. These limitations have been interpreted differently in similar studies, however the restrictions adopted in this research have been assumed by the following references [11,17,26]. Thereafter, two additional topographical factors were assumed, namely the length-to-height ratio and the terrain slope, regarded in almost all PHES evaluation studies. For the purpose of this investigation, thresholds of 20 [8,26] and

Table 2

Model specifications and restrictions areas data.

Criteria	Value
Minimum gross head	100 m
Maximum gross head	800 m
Maximum length-height ratio	20
Maximum terrain slope	5°
Minimum water bodies volume	1,000,000 m ³
Minimum distance from potential reservoir to roads and railways	200 m
Minimum distance from potential reservoir to inhabited areas	500 m
Forests and protected zones	–

5° [12,15,25,27], respectively, were set, since higher values led to considerable losses and additional engineering-related investment expenditures, which account for a significant portion of the capital cost. Therefore, potential locations that fulfill the above requirements offer the second reservoir opportunity associated with the water bodies. Accordingly, these locations represent the theoretical transformation capacity of the PHES. Subsequently, and in order to exclude unsuitable areas and deduce the physical potential, numerous land use and environmental restrictions are applied to the theoretical potential. These include road and railway infrastructure as well as inhabited areas. Indeed, a prospective candidate for the second reservoir must be located outside these areas, with a buffer zone of 100 and 200 m respectively [15,25,27]. From an environmental standpoint, forests and protected areas have been excluded, assuming that these areas are given particular priority owing to their importance for the preservation of the environment.

Through the application of the different specifications and restrictions of the model (Table 2), the decision-making analysis can be applied by considering the resulting physical potential associated with the two topologies of interest: T2 and T4.

2.2. Multi-criteria analysis: Analytic Hierarchy Process method

Ensuring an appropriate and sound decision in light of a number of considerations involving multiple decision makers and divergent objectives, however, remains a major concern. These issues are highly relevant to the Multi-Criteria Decision Making (MCDM) framework, which describes various techniques developed to assist policy and decision makers in structuring relevant data, evaluating the consequences of various options, and exploring their perspectives and requirements to make the ideal decision [28]. These techniques have been widely and successfully applied to a variety of purposes, including renewable energy policy and planning. MCDM methods have proven their effectiveness in tackling a wide variety of challenges related to site assessment studies incorporating renewable energy systems [29–33], while considering binding criteria in a structured manner.

In line with the literature, MCDM approaches may be divided into two categories: Multi-Objective Decision Making (MODM) and Multi-Attribute Decision Making (MADM). However, the only distinction between them is that MODM deals with a continuous decision domain, seeking to find an optimal solution within a set of predefined constraints. In contrast, MADM deals with a discrete domain of alternatives that are already predetermined [22,28]. There are various MADM techniques that are widely considered in the literature and have gained widespread use in renewable energy projects, including the following: TOPSIS, ELECTRE... and AHP [28]. Having proven its usefulness, the Analytic Hierarchy Process (AHP) method - primarily developed by Saaty [34] - has been adopted for the sake of this investigation due to its many appealing advantages, namely, its ability to help process complex, unstructured, multi-attribute queries, as well as its ease for decision makers to split a relevant problem into smaller, more manageable substructures. Indeed, the AHP incorporates the hierarchical component approach to facilitate decision-making, allowing the best possible decision among several crucial alternatives. The AHP method builds on the

pairwise comparison scheme to assign weights to each criterion. The pair-wise comparison matrix (A) for n criteria at a certain level can be constructed as indicated hereafter:

$$A = \{p_{ij}\}_{n \times n} = \begin{pmatrix} p_{11} & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ p_{n1} & \cdots & p_{nn} \end{pmatrix} = \begin{pmatrix} 1 & \cdots & p_{1n} \\ \vdots & \ddots & \vdots \\ \frac{1}{p_{1n}} & \cdots & 1 \end{pmatrix}$$

The presented matrix (A) is of size (n × n) wherein p_{ij} denotes the relevance of criterion i with respect to criterion j. The relative relevance is described using a numerical scale from 1 to 9 as detailed in Table 3 [34,35]. It is assumed that $p_{ii} = 1$, and $p_{ji} = 1/p_{ij}$, $\forall i \neq j$ for $i, j = 1, 2, \dots, n$.

The calculation of the weights forms the subsequent process, which consists in normalizing the matrix (A) in a couple of steps. First, summing the values of each column, then dividing each entry by the sum of its column (Eq. (1)). Finally, the relative weights are calculated by averaging the rows as presented in Eq. (2).

$$p_{ij}^* = \frac{p_{ij}}{\sum_{i=1}^n p_{ij}} \quad (1)$$

$$w_i = \frac{\sum_{j=1}^n p_{ij}^*}{n} \quad (2)$$

Subsequently, and in order to both verify and validate the consistency of the decision makers' pairwise comparison judgments, the Consistency Index (CI) (Eq. (3)) is first derived by calculating the eigenvector and maximum eigenvalue of each matrix before determining whether consistency is met by the Consistency Ratio (CR) calculated using Eq. (4). These two consistency metrics are outlined in the following:

$$CI = (\lambda_{max} - n) / (n - 1) \quad (3)$$

where λ_{max} stands for the maximum eigenvalue for each matrix and n refers to the number of criteria.

$$CR = CI / RI \quad (4)$$

where RI is the random consistency index, whose value is selected based on the dimension (n) of the comparison matrix (A). The values of RI are listed in Table 4.

- If $CR \leq 0.10$, the degree of consistency is fulfilling.
- If $CR > 0.10$, significant inconsistencies exist. In such a case, the pairwise comparison should be reconducted.

Table 3
The AHP method's pairwise comparison scale [34].

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Moderate importance	Experience and judgment strongly favor one activity over another
5	Essential or strong importance	Experience and judgment strongly favor one activity over another
7	Very strong importance	An activity is strongly favored and its dominance demonstrated in practice
9	Extreme importance	The evidence favoring one activity over another is of the highest possible order of affirmation.
2, 4, 6, 8	Intermediate values between the two adjacent judgments.	When compromise is needed.

In view of the multi-attribute nature of the present study, the AHP method was considered as a comprehensive approach to achieve the objective of evaluating and classifying the appropriate locations for the installation of PHES plants in two different topologies T2 and T4. This kind of methodology relies initially on the identification of the problem and the specific objective, as described above for the present case. Then, the different criteria and decision factors (sub-criteria) are defined, based on the experts and the literature. Afterwards, the alternatives, represented as sites (pixels), are established as the last hierarchical level. Thereafter, the pairwise comparison took place and the decision matrix was set up accordingly. The multiple AHP calculations described above are then performed, and the consistency of the decision is checked by means of the CR. Thus, overall weight is then concluded if consistency is met, otherwise the pairwise comparison must be reworked. Fig. 2 illustrates the presented hierarchical structure as well as the whole algorithm formulation of the AHP approach.

2.3. Definition of the decision criteria

A decision-making problem is approached by determining the various criteria upon which locations to install a PHES system will be evaluated. These decision criteria are established on the basis of studies conducted in related fields, expert judgments, and the authors' perspective by introducing two novel assessment factors that were considered relevant for the selection of suitable locations. The criteria set up cover different aspects affecting the technical, economic as well as environmental dimension, namely: geography, accessibility, risk and environment. These are listed in the following manner.

2.3.1. Geography

In terms of geography, the selected criteria include gross head, water body volume (in the case of the T2), length-to-height ratio, as well as terrain slope. These factors have been grouped together in the geographic dimension since they are the fundamental requirements for the selection of the PHES system. In subsequent sections, each of these factors will be detailed.

2.3.1.1. Gross head. Given that the hydroelectricity produced by the water flowing through the turbines is linearly related to the gross head (Eq. (5)), this criterion is taken into account in order to ensure the siting of PHES plants with advantageous gross heads and, consequently, maximizing the electricity produced on site. The gross head represents the difference in elevation between the two reservoirs, denoted by a letter H in Fig. 1. Beyond this engineering advantage, this factor also has a significant economic impact. Indeed, large differences in elevation can limit the volume of stored water while maintaining electrical power and, therefore, reducing related investment costs [8,37]. Therefore, locations with higher gross head are more attractive than others.

2.3.1.2. Length-to-height ratio. The length-to-height ratio (L/H) refers to the ratio of the horizontal distance to the gross head as separating two relevant reservoirs in the case of conventional-PHES or between the sea and the upper reservoir for the seawater-PHES, represented by L in Fig. 1. When the penstock or tunnel length is too long, not only the investment cost is high, but also the construction cost, and the resistance of the water supply is important, which directly leads to head losses [38,39]. Therefore, the length of the pipes should be as short as possible. The longer the distance (L), the more expensive the penstock construction. Thus, sites with a relatively small L/H ratio between the reservoirs are potentially attractive over others.

2.3.1.3. Terrain slope. A criterion as terrain slope is considered as a decision factor due to its remarkable impact on the project's investment cost, as most of it, especially for large reservoir plants, is generally associated with significant civil works, notably earthworks [40]. Thus, it

Table 4
RI values [36].

n	1	2	3	4	5	6	7	8	9	10
RI	0	0	0.58	0.90	1.12	1.24	1.32	1.42	1.45	1.49

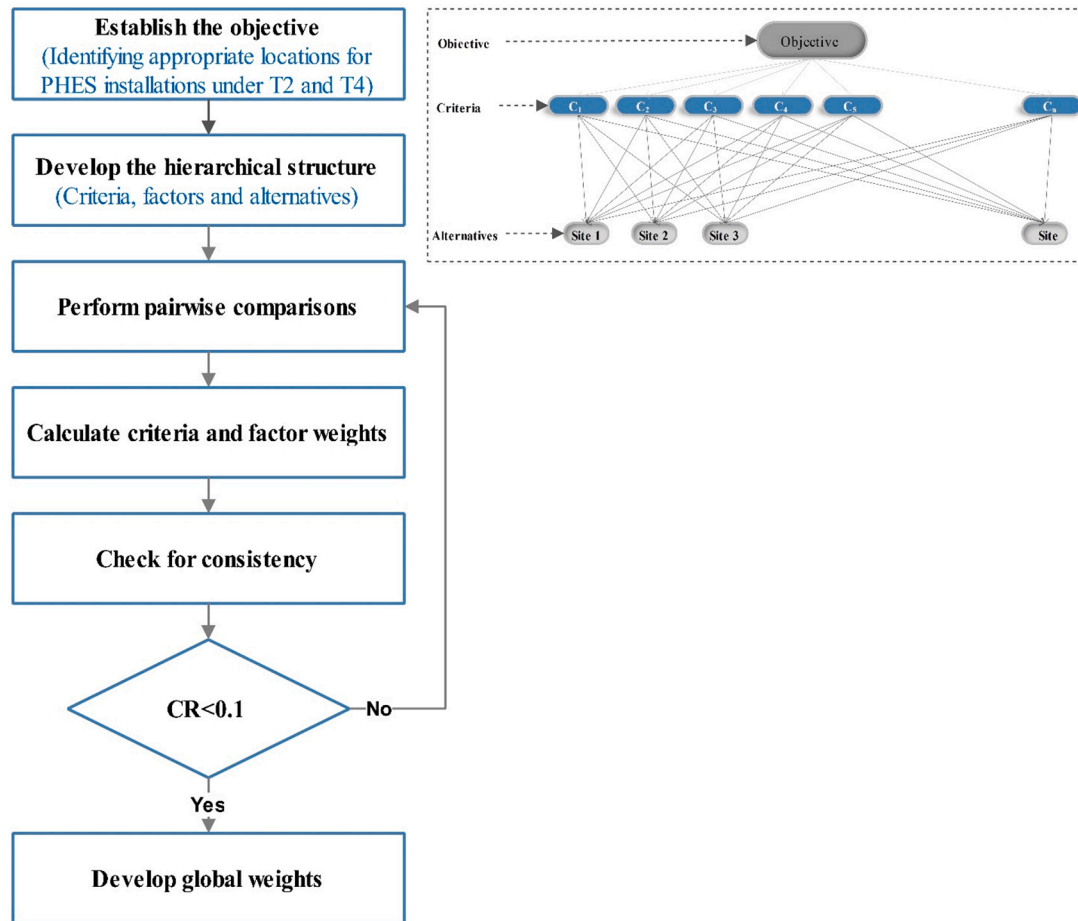


Fig. 2. AHP hierarchical structure and algorithm.

is worthwhile considering sites that are relatively flat and where it would be possible to build a reservoir, as it may be difficult for practical reasons to build a reservoir on steeply sloping sites [12,27]. Therefore, sites with slight slopes are more strongly preferred than others.

2.3.1.4. Water bodies volume. As water resources are critical to the development of PHES plants, the amount of water available in existing water bodies (reservoirs, dams, and lakes) is a key factor in providing a reliable assessment. As this factor is linearly related to stored energy, higher power generation is provided by higher water flow. Therefore, assuming a design that restricts the potential reservoir's maximum volume to that of the existing water bodies, as long as the prospective reservoir has a larger storage volume than the existing resource, sites in proximity to a high-capacity water resource are given priority over others [25]. This consideration, however, is not taken into account for seawater-PHES systems, since seawater is abundant.

2.3.2. Accessibility

With respect to accessibility, distance to transmission lines, roads and railways as well as urban areas were included within the decision criteria. These factors have a significant impact on the overall investment cost of the PHES plant, accounting for a significant share of it [40]. In this section, each of these parameters is described in detail.

2.3.2.1. Distance to transmission lines. After generating hydroelectricity by moving water through a turbine connected to an electrical generator, it will be relayed to the stations through the transmission lines. Therefore, the new PHES development needs to be located near existing power lines in order to reduce the initial investment and decrease the losses associated with the energy transmission [8,10,39,41]. Since the cost involved in building a new transmission line is substantial and could be complicated as it entails securing the necessary permits as well as land acquisition [38]. As a matter of fact, D. DeSantis et al. [42] reported that, according to the Black & Veatch cost model, the cost of a transmission line is \$ 1.42 M/km for >16 km of power lines, incorporating the costs of the transmission cable and towers. Therefore, the more conveniently the PHES is located near existing transmission lines, the lower the cost of grid integration will be. Consequently, sites close to power grids are more cost-effective than others.

2.3.2.2. Distance to roads and railways. The requirement for proximity to road infrastructure is contemplated as the construction of a PHES entails access to the site, maintenance and transportation of the equipment [10]. Since the cost associated with a new connection may increase the total project investment [40]. Consequently, locations in close vicinity to the existing roads and railroads are more cost effective and therefore more conducive than others.

2.3.2.3. Distance to inhabited areas. Incorporating the distance to urban areas into selection criteria ensures the PHES plant is feeding the generated electricity into the local demand and preventing the corresponding transmission costs [16]. Therefore, the closer the sites are to urban areas the more promising they will be.

2.3.3. Risk: seismic activity

With regards to the risk aspect, a PHES facility ought to be deployed in areas with favorable geological conditions in order to promote proper engineering and placement of the required equipment [38]. Therefore, the seismic activity criterion is taken into consideration in order to ensure that the construction of the upper reservoir of the plant is done where earthquakes are less frequent [16]. This factor is quite crucial, especially when prospecting for sites in coastal areas, to form a seawater-PHES plant, which are very sensitive to seismic activity [19]. Regardless of the technical impact, placing a PHES plant in zones with seismic vibrations designates additional costs induced by civil works such as weirs, waterways, tunnels, etc. [10]. Hence, as a consequence of this, sites with a lower risk of seismic activity are more suitable for development than others.

2.3.4. Climate

As for the climatic aspect, various criteria are involved, in order to explore the different factors that could affect the installation of the PHES plant and/or its operation and efficiency, the rainfall rate, the evapotranspiration, and also the sea surface salinity are considered for this study. These last two factors are used for the first time in similar studies seeking to prioritize appropriate sites with the lowest levels of these two additional decision parameters, thereby ensuring that the selected sites are cost-effective for further deployment of the studied systems. The evaluation of the climatic aspect of conventional PHES (T2) is based on the criteria of evapotranspiration and precipitation rate while the sea surface salinity is regarded for the seawater-PHES (T4). All of these factors are detailed hereafter.

2.3.4.1. Evapotranspiration. The so-called evapotranspiration (ET) phenomenon is the combination of two different processes: on one hand, the loss of water from the soil surface by evaporation and, on the other hand, the loss of water from the crop by transpiration [43]. Incorporating such hydrological parameter as evapotranspiration would further improve the effectiveness of the present work model. Such parameter would in fact prove useful to deduce the probable evaporation losses from the storage reservoirs, especially in arid and/or semi-arid regions such as the case under study [44]. In these regions, in fact, studies have reported that about 40 % to up to 50 % of the stored water reservoir capacity is lost through evaporation [45,46]. This kind of parameter along with other geological parameters is necessary when evaluating the hydroelectric energy potential [47,48]. In this respect, the evapotranspiration criterion was taken into account for the first time among the decision criteria in previous similar studies on the evaluation of the potential of the PHES facilities. Thus, locations with relatively lower ET values are more attractive compared to others.

2.3.4.2. Rainfall. As for the rainfall rate, it is regarded as a decision criterion in order to compensate the losses due to evapotranspiration and leakage [11]. Since in many locations, there is an approximate equilibrium between annual precipitation and evaporation over the reservoir [49]. There are two other ways in which rainfall can be effective: rainwater harvesting to increase water storage and reducing pumping requirements [8]. Such a parameter is crucial when it comes to evaluating the hydropower energy potential and thereby ensuring a reliable assessment [47]. Sites receiving higher precipitation are therefore more favorable than others.

2.3.4.3. Sea surface salinity. Sea Surface Salinity (SSS) is one of the key

climate variables recognized by the Global Climate Observing System (GCOS). Indeed, it is widely referred to as the concentration of salt in seawater. The development of seawater PHES, which use the sea as a lower reservoir by storing seawater, has brought a certain difficulty such as the impact of salt water leakage on the surrounding area and the corrosive effect on the associated equipment [50]. The establishment of this kind of system indeed presents a significant challenge owing to the corrosive effect of seawater on the pump turbines. Nevertheless, it has been proven that the adoption of anticorrosive materials for seawater pump turbines and penstocks has reduced the restrictions on the location of seawater-PHES plants [51,52]. The incorporation of such a parameter - sea surface salinity - in the decision-making process, used for the first time in similar studies exploring suitable sites for the installation of PHES power plants, prevents the placement of a power plant in sites with high salinity, which leads to a significant corrosion effect and thus to an additional investment and a shorter plant lifetime. As the study area is part of the high sea surface salinity regions, the introduction of the SSS is nevertheless undertaken with the aim of favoring low salinity sites over others, as the present study area comprises the Atlantic and the Mediterranean, which have distinct salinity levels.

Based on the located and identified physical potential, the different criteria, factors referred to for both topologies: T2 and T4, are ranked. These different selection criteria are subjected to a decision analysis based on their priority level, where a weight is assigned to each criterion and factor as per the AHP methodology described in Section 2.2. The established priority orders are presented in the results and discussion section, where the approach outlined earlier is extended to the Moroccan context - the case study. As a follow-up to the comprehensive approach, the screening criteria data are prepared into raster data through a collection of GIS tools in ArcToolbox. Subsequently, using the ModelBuilder feature, the criteria and factors related to each of the two topologies and their respective weights were used to generate the site ranking maps depicting the classification of the various locations from suitable to excellent. Fig. 3, summarizes holistically the workflow structure undertaken throughout this study.

2.4. Assessment of potential storage capacity

The potential energy stored in the prospective locations - secondary reservoirs - for the two topologies: T2 and T4 is given by the following equation:

$$E = \rho \bullet g \bullet H \bullet V \bullet \eta / 3600 \quad (5)$$

where, E is the potential energy storage capacity in Wh, ρ is the water density deemed to be 1019 kg/m³, g is the gravitational acceleration assumed to have a value of 9.81 m/s², H is the gross head in m, V is the water volume within the second reservoir in m³, and η is the turbine conversion efficiency assumed to be 90 % [11,15,27,41,49,53].

In terms of the estimation of the volume stored in the second reservoir, the approach undertaken by N. Fitzgerald et al. [27] and N. Ghorbani et al. [15] was also regarded in this study. Thus, a depth of 20 m along with a minimum surface area of 70,000 m² ought to be provided for the development of the PHES facilities as per T2 and T4, while allowing around 28.57 % of these areas for civil engineering works.

3. Application of the methodology outlined: case study of Morocco

3.1. Background

Geographically, Morocco is positioned in northwest Africa, overlooking the Mediterranean Sea on the north and the Atlantic Ocean on the west (Fig. 4). The country covers an overall surface area of 710,850 km², of which a large part is covered by mountainous areas. The main mountain ranges include the Rif, the Middle and High Atlas Mountains,

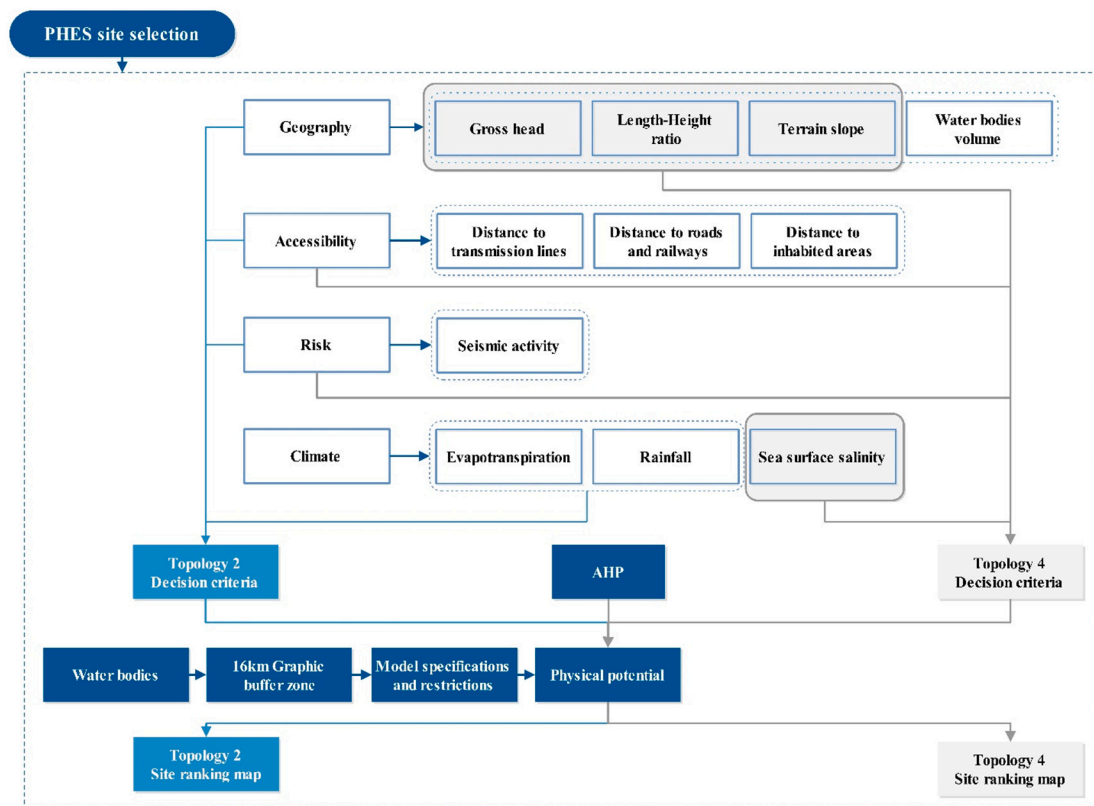


Fig. 3. PHES site selection workflow chart intended for T2 and T4.

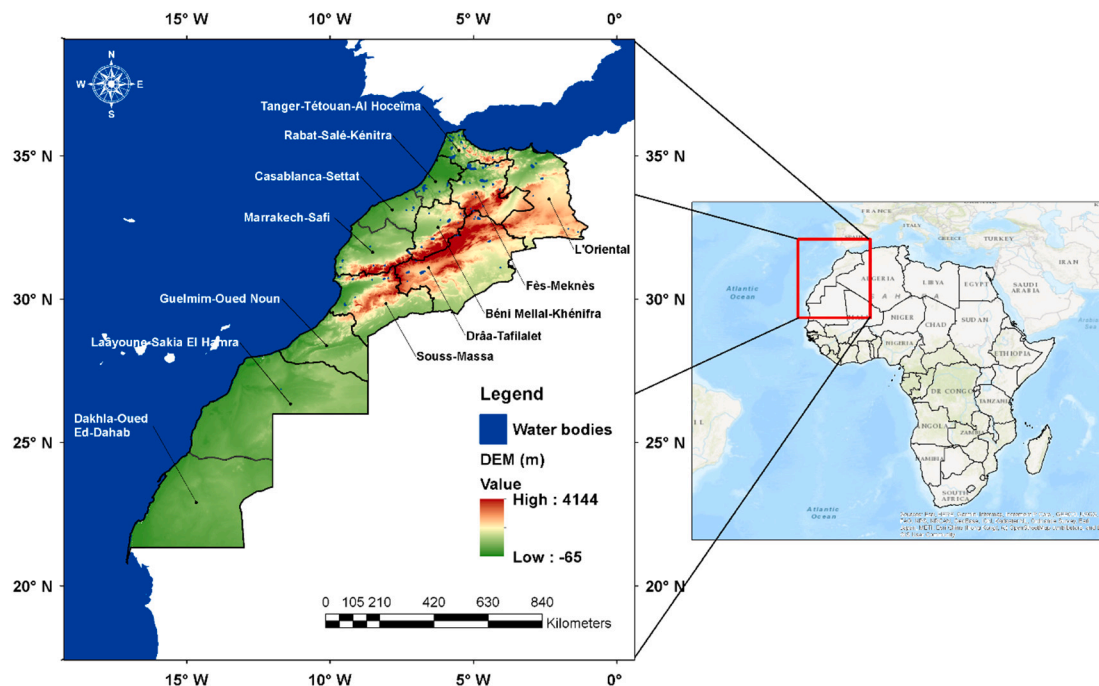


Fig. 4. Morocco: geographical position.

which reach heights ranging from 2000 to over 4000 m [54]. In 2019, Morocco's population reached 35,586,616 placing it as the 11th most populous African nation, with the largest share concentrated in the Casablanca-Settat region, deemed the most dynamic nationwide [55].

Regarding the energy context, the kingdom is heavily dependent on fossil fuels, which accounted for 93.3 % in 2016 [56]. Enjoying the

substantial renewable energy resources, with irradiation levels of >2300 kWh/m²/year, and average wind speeds of >6 m/s on the Atlantic coast [57], the country launched a strategy in 2009 to promote the development of renewable energy. In fact, this policy was initiated with a 42 % target share in 2020 and then strengthened through a new objective within the framework of the COP21, i.e. to increase the national

renewable energy mix to 52 % by 2030: 20 % solar energy, 20 % wind energy and 12 % hydropower [58]. This strategy intends to reduce greenhouse gas (GHG) emissions and decrease energy dependency so as to ensure environmental protection and energy security. The development of hydroelectricity was initiated in the 1960s with a policy of building large dams to strengthen the hydroelectric portfolio. This energy has always been seen as a component of the country's power plant mix. However, hydropower generation is highly dependent on rainfall levels, which makes it difficult to predict. Nevertheless, the country is targeting smaller-scale projects as well as the expansion of innovative technologies such as PHES plants to increase hydropower capacity to 3 GW by 2030 [59]. Within the framework of the national policy, this study identifies suitable sites for the development of PHES plants, applying the methodology detailed in Section 2 according to two topologies: (1) T2 by taking advantage of existing water bodies: reservoirs, dams and lakes. This kind of system was introduced with the first power plant of Afourar with a capacity of 464 MW put into service in 2004 [58]; (2) T4 by exploring a coastline extending over 3500 km on two maritime sides: Atlantic (2934 km), and Mediterranean (512 km). Nonetheless, this kind of technology has not yet been deployed in Morocco. The adoption of PHES systems will contribute to increase the share of intermittent renewable energies - in particular SPV and wind - thanks to the storage provided by these systems.

3.2. Data collection and processing and overall process workflow

Having established the methodology, the appropriate data for the present case study is prepared and/or retrieved and adapted to the raster data format, before proceeding to generate the ranking maps using the weights assigned by the AHP method and the tools and features of ArcGIS 10.8 software. The main steps followed during the data collection and processing phase include the following:

(1st): Build a 16 km buffer zone around water resources with storage volume greater than $1,000,000 \text{ m}^3$ - for T2, nevertheless, the buffer zone covers the entire coastal area - for T4 - thanks to the “Graphic buffer” tool (Fig. 5(a) and (b)). The water bodies database used in this study was retrieved from the Geofabrik website which provides OpenStreetMap data [60], where the majority of the identified areas are located in the northern part of the country while the south is covered by the desert. However, the extracted data on water bodies do not provide information on their volume and elevation.

Therefore, a digital elevation model (DEM) is required to assess them. The DEM data used were obtained from NASA's Shuttle Radar Topographic Mission (SRTM) with a resolution of 3 arcsec (90 m) (Fig. 6(a)) [61]. The approach involves leveraging the surrounding topography to estimate the storage volume, which is computationally efficient. In order to do so, a tool referred to as “Polygon Volume” was adopted using Triangular Irregular Networks (TIN) created from the DEM. In addition, the “zonal statistics as table form” tool was used to get the average elevation of each of the water bodies.

(2nd): Following the volume calculation, the various necessary data related to the model restrictions and specifications detailed in Section 2.1 are prepared in the required format in order to identify the theoretical and physical potential of the two topologies under study: T2 and T4. Indeed, the slope in the 16 km buffer zone was calculated using “slope” tool (Fig. 6(b)). Similarly, the gross head was deduced by subtracting the difference between the pixel DEM values of the sites in the vicinity of the water body and its determined elevation, by means of a created grid table having the same mesh resolution as the DEM used. For the length-to-height ratio, the “Near” tool derived the closest distance between the water body and its surrounding pixels, and then the ratio between the latter and the gross head was deduced. As for the transmission lines, roads, railways, inhabited areas (Fig. 6(c), (d) and (e)), forests and protected zones, they were obtained from [60]. Given the restrictions defined, the “buffer” tool was adopted to generate the limits specified in the methodology for these areas. Having identified the areas deemed appropriate, and in order to conduct the spatial decision based on the AHP process, remaining data related to the established factors are derived as follows: (a) the seismic activity data was digitized from the seismic zoning map adopted by the Moroccan regulation on earthquake-resistant construction (RPS 2000) (Fig. 6(f)) [62]. (b) Evapotranspiration data was retrieved from the Global Aridity Index and Potential Evapotranspiration climate, providing high-resolution annual average ET0 data (Fig. 6(g)) [63]. (c) Precipitation data were obtained from the Center for Hydrometeorology and Remote Sensing (CHRS) website [64]. And in order to guarantee a reliable assessment, a 10-year average was calculated accordingly (Fig. 6(h)). (d) Sea surface salinity information was provided by NASA's Soil Moisture Active Passive (SMAP) observatory, which is produced operationally by Remote Sensing Systems (RSS) with a latency of one month (Fig. 6(i)) [65].

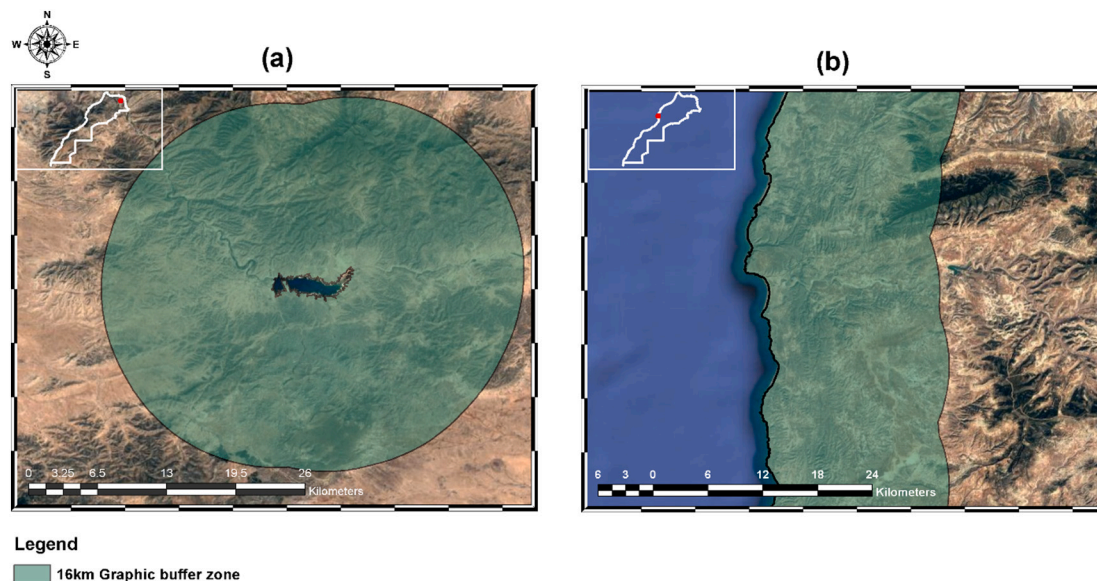


Fig. 5. 16 km graphic buffer: (a) T2; (b) T4.

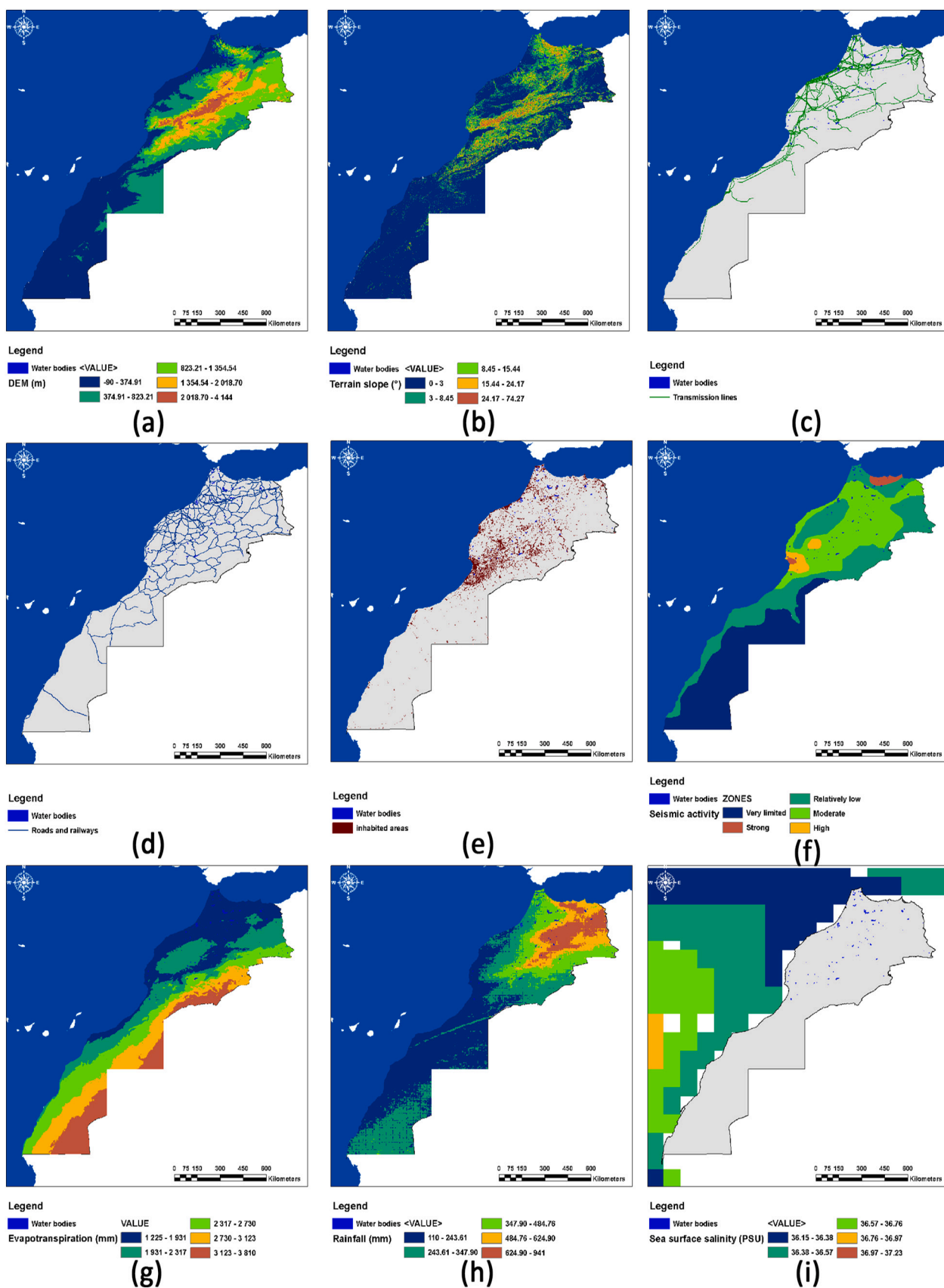


Fig. 6. Decision criteria data: (a): DEM, (b): terrain slope, (c): transmission lines, (d): roads and railways, (e): Inhabited areas, (f) seismic activity, (g):evapotranspiration, (h): rainfall, (i): sea surface salinity.

(3rd): In view of the different data processed and collected, and in order to generate the ranking maps of favorable sites for the two PHES topologies of interest, a classification process based on expert judgments and literature results was performed. Table 5 summarizes the various classification categories for T2 and T4, prioritizing locations with attractive values over those without. Then, taking into account the AHP weights deduced from the method processing, the ranges of criteria collected, the “Raster calculator” tool as well as the ModelBuilder feature, the spatial decision was conducted and the two ranking maps were generated accordingly.

4. Results and discussion

Within the presented study area - Morocco - suitable locations that meet the requirements of the approach presented in this paper are identified, offering the possibility to develop new storage reservoir(s) in an optimal topographic location. This assessment is based on one of the most used combination in the energy planning field, which is the multi-criteria analysis presented by the AHP method along with the GIS tools. In fact, the physical potential - realizable - has been identified in the first

place before being subjected to a decision analysis that allows defining the areas with the highest weight through the resulting promising site ranking maps. Sections 4.1 and 4.2 outline the different findings for each of the two studied topologies, the conventional-PHES - T2 - and the seawater-PHES - T4, respectively.

4.1. Potential sites under T2

The assessment of the PHES potential under T2 in Morocco has been carried out by taking advantage of its existing dams and lakes to form a conventional PHES. Indeed, 88 reservoirs have been retained with a capacity $>1,000,000 \text{ m}^3$. Besides, the altitude chart of these reservoirs reveals that a large part of them is located between 100 and 400 m of height above mean sea level (AMSL). Fig. 7(a) presents the altitude level of the selected water bodies. Across the defined 16 km buffer, the realizable potential has been identified considering the various model specifications and restrictions stated in the methodology, the corresponding data collected and processed, and the ArcGIS geoprocessing tools. Fig. 7(b) illustrates the various limitations within the area surrounded by one of the water bodies considered. Indeed, merely 1.40 % of the overall buffer area is deemed adequate for the placement of an upper or lower reservoir. The promising sites were then refined based on the $70,000 \text{ m}^2$ threshold area requirement for a PHES under T2, as described in Section 2.4. As a result, 936 sites were identified across 9 of the 12 administrative regions nationwide, with an absolute average gross head of approximately 352 m. These sites offer an overall storage capacity of about 5403 million m^3 corresponding to a total prospective energy storage capacity of about 4931 GWh. At the regional level, the Beni Mellal-Khénifra and Fez-Meknes areas have the largest share, i.e. 42.8 % of the total potential sites. Table 6, presents the details regarding the candidate sites available for each region, their corresponding total storage volume and their total energy storage capacity. In terms of the latter, the Rabat-Salé-Kénitra region joins Beni Mellal-Khénifra in having almost the same amount of storage, accounting for 39.4 % of the total available volume. However, the potential energy storage capacity of the Rabat-Salé-Kénitra region is only 28.2 % of that offered by Beni Mellal-Khénifra due to the fact that its absolute average gross head is more than three times that of Rabat-Salé-Kénitra. Indeed, the gross head is directly related to the potential energy storage capacity (Eq. (5)). On the basis of this criterion, the Beni Mellal-Khénifra region offers more promising sites with a more interesting gross head, as they provide an equivalent amount of energy with smaller reservoirs.

Subsequently, the multi-criteria analysis was conducted, using the AHP method, on the identified feasible potential to select the most advantageous candidate locations. A pairwise comparison was performed to assign weights to each criterion based on expert judgments, and literature findings. The resulting weights were approved by a CR that is approximately 3.4 %, below the 10 % threshold set by the method adopted (Section 2.2). Once the consistency was fulfilled and in agreement with the AHP algorithm, the overall weights were concluded and are highlighted in Table 7, where the geographic factor accounts for the lion's share, with 59.10 %. Indeed, the geography of the site is a relevant criterion, as convenient conditions increase the performance of the scheme and reduce the required investments with higher gross heads and storage volumes, as well as lower length-to-height ratios and terrain slopes. In terms of accessibility, a ratio of 21.79 % has been assigned, split between the different infrastructures, with accessibility to transmission lines receiving the greatest weight, i.e. 11.81 %. In fact, the cost of a new power line is significant and the losses incurred by the transport of electricity when the plant is placed far from the grid. Therefore, this factor was prioritized over other forms of infrastructure. With respect to the risk criterion presented by seismic activity, a weight of 12.05 % was assigned, placed in third place due to its multiple impacts on the system (as stated in Section 2.3.3). Thus, sites with little earthquake activity were favored over others. On the climate endpoint, a proportion of 7.06 % was accorded, nearly equally divided between the precipitation and

Table 5

Range-based prioritization of the criteria and factors adopted for the T2 and T4 PHES classification.

Criteria	Factor	Unit	Ranges for T2	Ranges for T4
Geography	Gross head	m	≥ 300	≥ 300
			200–300	200–300
			150–200	150–200
			100–150	100–150
	Length-height ratio	–	≤ 5	≤ 5
			5–10	5–10
			10–15	10–15
			15–20	15–20
	Terrain slope	°	≤ 3.5	≤ 3.5
			3.5–4	3.5–4
			4–4.5	4–4.5
			4.5–5	4.5–5
Accessibility	Water bodies volume	Million m^3	≥ 100	–
			10–100	–
			5–10	–
			1–5	–
	Distance to transmission lines	km	≤ 1	≤ 1
			1–5	1–5
			5–10	5–10
			>10	>10
	Distance to roads and railways (km)	km	≤ 1.5	≤ 1.5
			1.5–5	1.5–5
			5–7.5	5–7.5
			>7.5	>7.5
Risk	Distance to inhabited areas	km	≤ 1	≤ 1
			1–5	1–5
			5–20	5–20
			>20	>20
	Seismic activity	–	Low risk (1 & 2)	Low risk (1 & 2)
			Medium risk (3)	Medium risk (3)
			High risk (4)	High risk (4)
			Extremely high risk (5)	Extremely high risk (5)
Climate	Evapotranspiration	mm	≤ 2000	–
			2000–2500	–
			2500–3000	–
			>3000	–
	Rainfall	mm	≥ 550	–
			350–550	–
			250–350	–
			<250	–
	Sea surface salinity	PSU	–	≤ 36.15
			–	36.15–36.38
			–	36.38–37.00
			–	>37

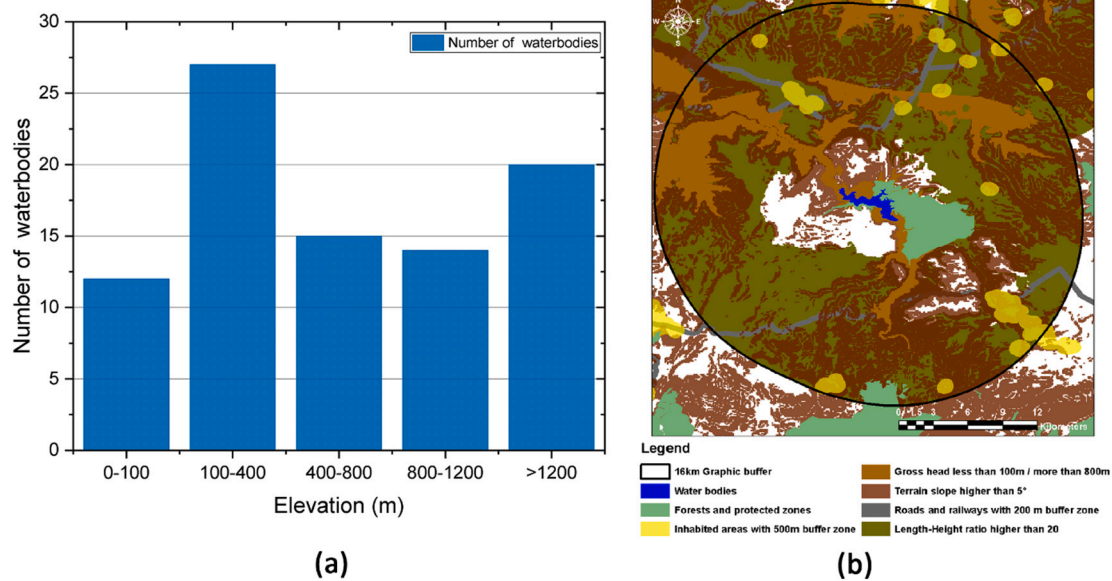


Fig. 7. (a): Elevation chart of selected water bodies with a storage capacity equal to or $>1,000,000 \text{ m}^3$, (b): model specifications and restrictions.

Table 6

Number of potential sites, volume, absolute average gross head and energy storage capacity of PHES plants under T2 for the 12 regions of Morocco.

Administrative region	Number of potential sites	Total potential storage volume (million m^3)	Absolute average gross head (m)	Potential energy storage capacity (GWh)		
				Min ^a	Max ^b	Sum total
Béni Mellal-Khénifra	206.00	1248.10	522.31	0.29	223.08	1861.24
Casablanca-Settat	43.00	310.28	170.11	0.38	32.54	140.80
Drâa-Tafilalet	125.00	692.17	376.36	0.30	63.17	600.79
Fès-Meknès	195.00	880.42	301.56	0.26	57.20	648.64
Marrakech-Safi	92.00	459.53	421.11	0.32	48.52	500.93
L'Oriental	45.00	303.23	378.86	0.30	203.59	405.68
Rabat-Salé-Kénitra	150.00	1299.14	150.23	0.26	40.35	525.80
Souss-Massa	46.00	105.32	415.11	0.37	11.58	120.14
Tanger-Tétouan-Al Hoceïma	34.00	104.59	409.12	0.30	21.20	126.88
Dakhla-Oued Ed-Dahab	0.00	0.00	—	0.00	0.00	0.00
Guelmim-Oued Noun	0.00	0.00	—	0.00	0.00	0.00
Laâyoune-Sakia El Hamra	0.00	0.00	—	0.00	0.00	0.00
Total/average	936.00	5402.78	522.31			4899.10

^a The smallest site.

^b The biggest site.

Table 7

T2 criteria weights according to the AHP analysis.

Criteria	Weight (%)	Factor	Weight (%)
Geography	59.10	Gross head	26.73
		Length-height ratio	17.46
		Terrain slope	9.98
		Water bodies volume	4.92
		Distance to transmission lines	11.81
Accessibility	21.79	Distance to roads and railways	7.12
		Distance to inhabited areas	2.86
		Seismic activity	12.05
Risk	12.05	Evapotranspiration	3.67
Climate	7.06	Rainfall	3.39

evapotranspiration factors. The latter is considered for the first time in similar assessment studies, to our knowledge. Assuming that a certain equilibrium between ET_0 and precipitation rate needs to be guaranteed in the studied area, therefore the attractive locations are those with a certain balance between these two elements or those with higher precipitation than ET_0 .

As soon as the various factors involved for this topology were

evaluated through the AHP method, the GIS tools enabled the ranking and combination of the different layers (raster data) - in the framework of the identified feasible potential - using the different ArcToolbox tools as well as the ModelBuilder feature, with the assigned weights. As a result of this calculation, the site ranking map for T2 is generated. Fig. 8 depicts the outcome as six zones are presented, displaying the most promising sites. In fact, the weights range from a minimum of 0.42 to a maximum of 0.90, the latter value being the one that accumulates the best score from each of the ten factors. The approach undertaken revealed that the case study country presents numerous propitious sites for the development of this kind of topology, as 41.5 % of the identified locations were deemed to be excellent with a score higher than 0.70. Although the Beni Mellal-Khénifra region received the majority of excellent sites with around 31 %, the Fes-Meknes has nevertheless advantageous potential places with a maximum weight of 0.90. It would therefore be beneficial for investors and stakeholders to explore these beneficial spots for the development of PHES plant under T2. The various findings are summarized in Table 8.

With regard to the zones of Dakhla-Oued Ed-Dahab, Guelmim-Oued Noun, and Laâyoune-Sakia El Hamra, they are marked by the absence of potential locations for PHES since none of their existing reservoirs has

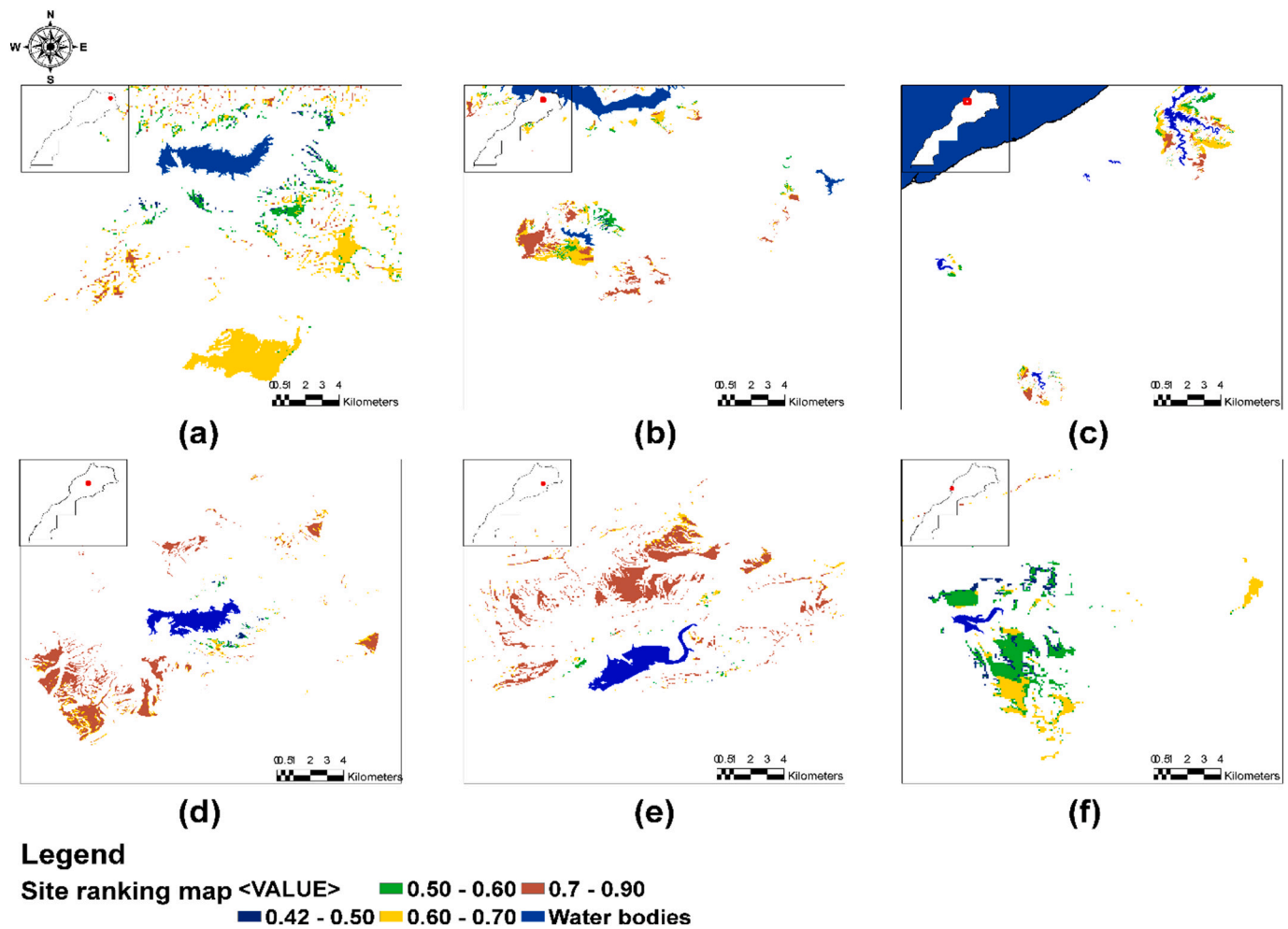


Fig. 8. PHES site ranking map under T2: (a) zone 1, (b) zone 2, (c) zone 3, (d) zone 4, (e) zone 5, (f) zone 6.

Table 8

Number of excellent sites and the assigned weight (min, mean, max) for the 12 regions of Morocco for PHES plants under T2.

Administrative region	Number of potential sites	Number of excellent sites (weight higher than 0.70)	Weight (%)		
			Min	Average	Max
Béni Mellal-Khénifra	206.00	122.00	0.48	0.71	0.86
Casablanca-Settat	43.00	14.00	0.54	0.66	0.81
Drâa-Tafilalet	125.00	43.00	0.46	0.68	0.88
Fès-Meknès	195.00	93.00	0.46	0.70	0.90
Marrakech-Safi	92.00	25.00	0.44	0.65	0.78
L'Oriental	45.00	14.00	0.52	0.68	0.80
Rabat-Salé-Kénitra	150.00	42.00	0.53	0.66	0.84
Souss-Massa	46.00	11.00	0.42	0.65	0.81
Tanger-Tétouan-Al Hoceïma	34.00	24.00	0.57	0.75	0.88
Dakhla-Oued Ed-Dahab	0.00	0.00	0.00	0.00	0.00
Guelmim-Oued Noun	0.00	0.00	0.00	0.00	0.00
Laâyoune-Sakia El Hamra	0.00	0.00	0.00	0.00	0.00
Total	936.00	388.00			

the threshold volume set in the methodology, regardless of the fact that they are mostly deserted.

4.2. Potential sites under T4

As far as this topology is concerned, the sea is used as a lower reservoir to form a PHES plant operating with saltwater instead of freshwater, given its particularity with respect to the conventional plant (T2). The developed model evaluates the entire Moroccan coastline topography - case study - which stretches over 3500 km on two sea shores: Atlantic and Mediterranean, that are 2934 km and 512 km respectively. Thereby, the realizable potential within the 16 km buffer zone has been identified similarly to T2, taking into account the various limitations of the model described in Section 2.1, the corresponding data collected and processed, and the ArcGIS geoprocessing tools. Fig. 9 illustrates the result obtained, on one of the considered coastal areas. In fact, merely 0.50 % of the entire buffer area is deemed adequate for the placement of an upper reservoir. The prospective sites were then threaded based on the deemed threshold of 70,000 m² surface area required for a PHES under T4. This led to a total of 670 sites being identified across 8 of the 9 regions covering part of the Moroccan littoral. Although the average gross head of these potential sites is about 211.20 m, the overall storage volume is about 4200 million m³. Thus, a total potential energy storage capacity of about 1931 GWh is available. From a regional perspective, the Marrakech-Safi and Souss-Massa area account for the largest share, nearly 51 % of the total identified potential sites (Table 9). Nevertheless, the Marrakech-Safi area offers only 11 %

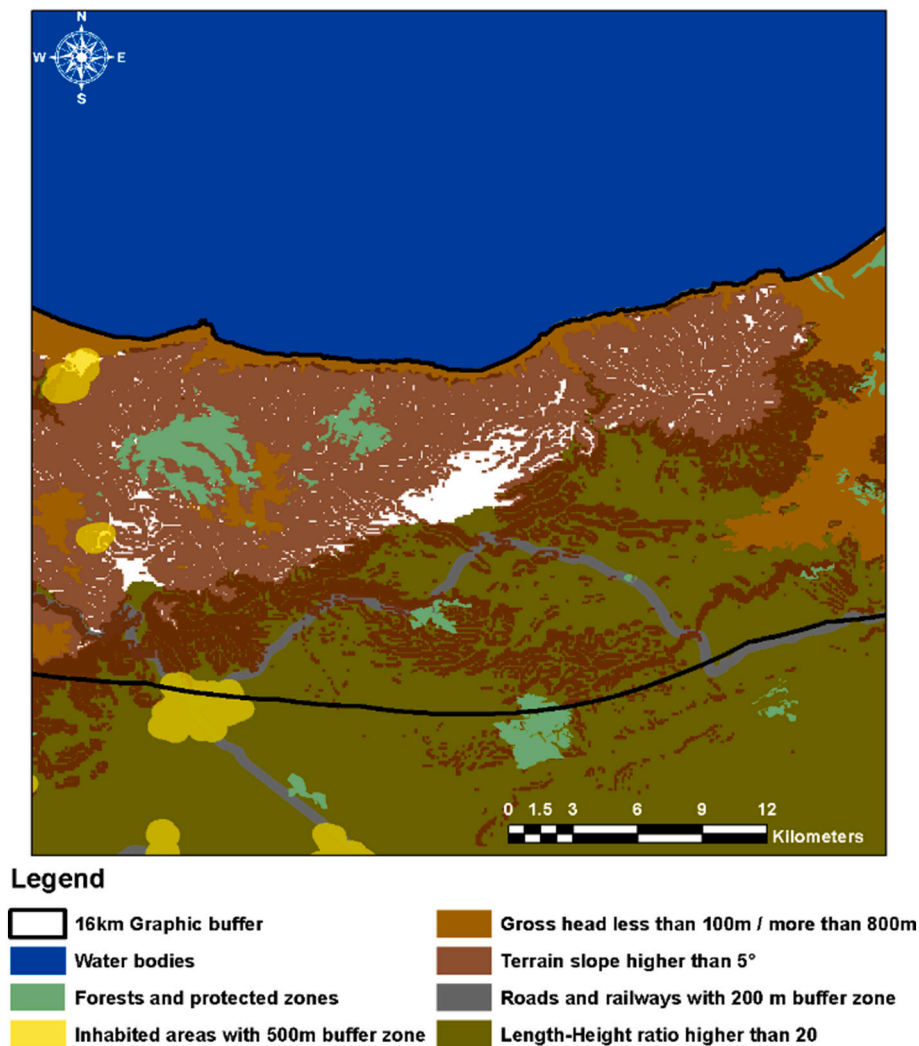


Fig. 9. Model specifications and restrictions for T4.

Table 9

Number of potential sites, volume, absolute average gross head and energy storage capacity of PHES plants under T4 for the 12 regions of Morocco.

Administrative region	Number of potential sites	Total potential storage volume (million m ³)	Absolute average gross head (m)	Potential energy storage capacity (GWh)		
				Min ^a	Max ^b	Total
Casablanca-Settat	10.00	43.12	115.84	0.25	3.34	11.35
Dakhla-Oued Ed-Dahab	8.00	43.36	105.80	0.21	2.99	10.31
Guelmim-Oued Noun	63.00	297.09	259.77	0.25	40.49	200.21
Laâyoune-Sakia El Hamra	122.00	1785.44	135.31	0.24	38.85	538.65
L'Oriental	96.00	345.13	287.05	0.24	64.87	240.40
Marrakech-Safi	180.00	952.84	236.40	0.25	28.80	484.85
Souss-Massa	160.00	620.34	287.73	0.01	16.65	374.00
Tanger-Tétouan-Al Hoceïma	31.00	112.38	261.69	0.34	23.53	71.56
Béni Mellal-Khénifra	0.00	0.00	–	0.00	0.00	0.00
Drâa-Tafilalet	0.00	0.00	–	0.00	0.00	0.00
Fès-Meknès	0.00	0.00	–	0.00	0.00	0.00
Rabat-Salé-Kénitra	0.00	0.00	–	0.00	0.00	0.00
Total/average	670.00	4199.71	211.20			1931.34

^a The smallest site.^b The biggest.

less energy storage potential than the Laâyoune-Sakia El Hamra area, which has twice as much storage capacity as Marrakech-Safi. This is due to the gap between these areas in terms of average head, such that the Marrakech-Safi region is almost twice that of the Laâyoune-Sakia El Hamra is. Indeed, the interesting gross heads maximize the amount of

energy supplied at these sites (Section 2.3.1). Given these results, the sites located in the Marrakech-Safi area are deemed to be the most promising.

Thereafter, and as with T2, an AHP analysis was undertaken to select the most beneficial and cost-effective candidate sites, given the physical

Table 10
T4 criteria weights according to the AHP analysis.

Criteria	Weight (%)	Factor	Weight (%)
Geography	59.10	Gross head	31.43
		Length-height ratio	18.74
		Terrain slope	8.93
Accessibility	21.79	Distance to transmission lines	11.81
		Distance to roads and railways	7.12
		Distance to inhabited areas	2.86
Risk	12.05	Seismic activity	12.05
Climate	7.06	Sea surface salinity	7.06

potential identified. Where a CR of 3.4 % approved the consistency of the analysis, the global assigned weights are conveniently summarized in Table 10. As for the climatic aspect, represented by the novel factor of sea surface salinity, considered for the first time in similar seawater-PHES studies, a weight of 7.06 % was allocated. In light of the fact that the development of seawater-PHES presents certain challenges such as the impact of saltwater leakage on the surrounding area and the corrosive effect on the related equipment (Section 2.3.4). Therefore, this factor was introduced in order to promote low salinity sites located mainly in the Mediterranean and northern shore of the Atlantic regions over those in the southern shore of the Mediterranean and more specifically the identified desert sites.

Subsequently, the GIS-based tools were implemented to combine the evaluation factor weights and the different layers prepared so as to generate the T4 site ranking map. Fig. 10 depicts the outcome as six areas are presented, highlighting the most favorable areas. Thus, the resulting scores reach a maximum of 0.84, covering the best part of the involved factors' areas. As a matter of fact, 30 % of the locations were

deemed highly suitable for the development of PHES for the T4 configuration, with a score above 0.65. Thus, this demonstrates that the country has a significant opportunity for policy makers to deploy this kind of system, which is currently unexplored in Morocco. Although the Marrakech-Safi and Souss-Massa regions accounted for the largest share

Table 11
Number of excellent sites and the assigned weight (min, mean, max) for the twelve regions of Morocco for PHES plants under T4.

Administrative region	Number of potential sites	Number of excellent sites (weight higher than 0.65)	Weight (%)		
			Min	Average	Max
Casablanca-Settat	10.00	0.00	0.42	0.49	0.54
Dakhla-Oued Ed-Dahab	8.00	0.00	0.46	0.53	0.60
Guelmim-Oued Noun	63.00	14.00	0.34	0.58	0.74
Laâyoune-Sakia El Hamra	122.00	22.00	0.41	0.59	0.72
L'Oriental	96.00	20.00	0.32	0.58	0.74
Marrakech-Safi	180.00	59.00	0.38	0.61	0.76
Souss-Massa	160.00	70.00	0.34	0.65	0.83
Tanger-Tétouan-Al Hoceïma	31.00	13.00	0.45	0.63	0.84
Béni Mellal-Khénifra	0.00	0.00	0.00	0.00	0.00
Drâa-Tafilalet	0.00	0.00	0.00	0.00	0.00
Fès-Meknès	0.00	0.00	0.00	0.00	0.00
Rabat-Salé-Kénitra	0.00	0.00	0.00	0.00	0.00
Total	670.00	198.00			

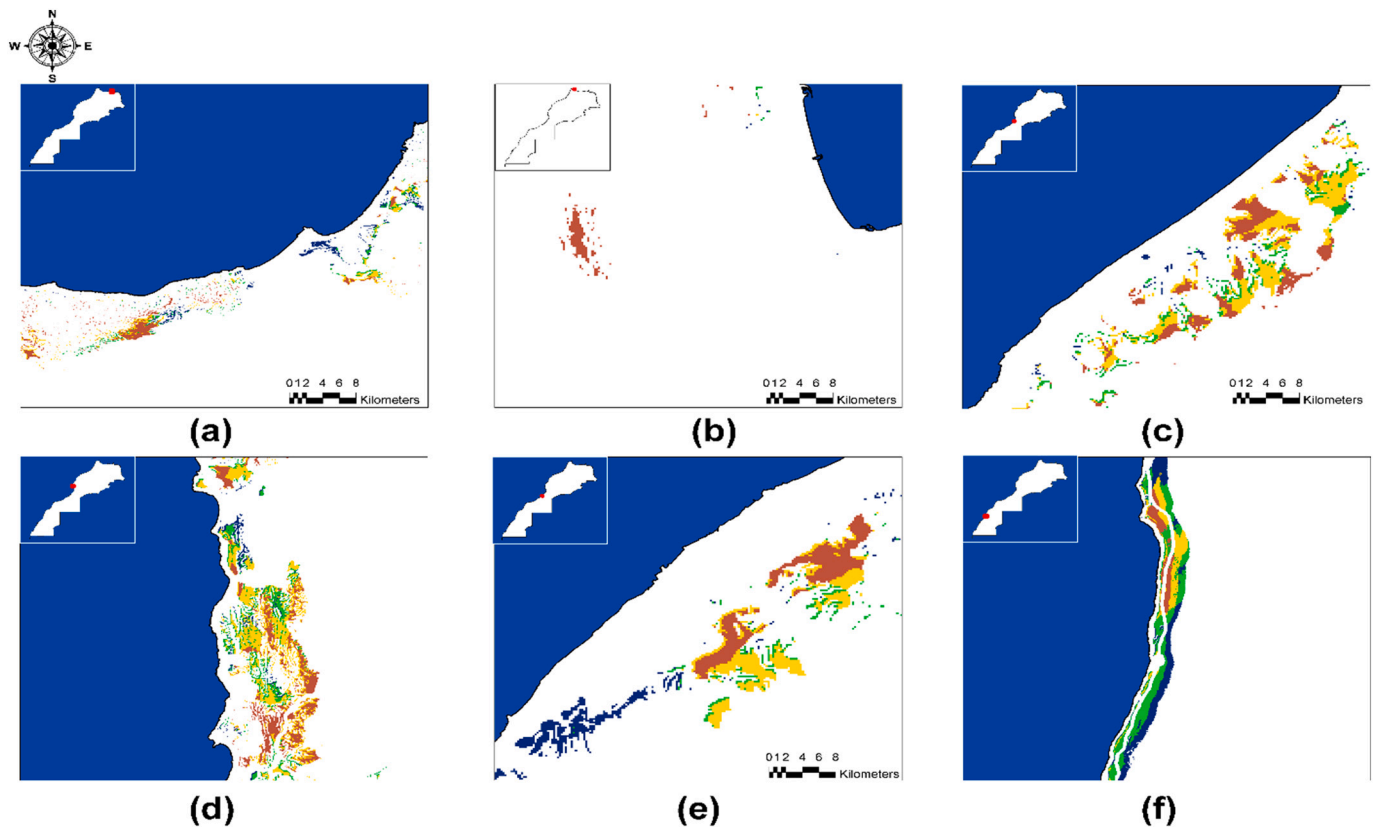


Fig. 10. PHES site ranking map under T4: (a) zone 1, (b) zone 2, (c) zone3, (d) zone 4, (e) zone 5, (f) zone 6.

of excellent sites with about 65.2 % in total, with beneficial candidate sites scoring above 0.76, along with the Tangier-Tetouan-Al Hoceima region. Table 11 summarizes these different findings.

As for the areas of Beni Mellal-Khenifra, Drâa-Tafilalet, and Fez-Meknes, they do not comprise a coastline, nevertheless, the area of Rabat-Salé-Kénitra does not meet the requirements and restrictions of the model and therefore no specific site has been detected.

On the basis of these findings, an analysis of the overall potential of the country studied - Morocco - in terms of PHES system under the two topologies (T2 & T4) considered, led to the following conclusions: (1) A total of 1606 sites were counted in the 12 administrative subdivisions, of which Marrakech - Safi is the leading one (16.94 %). (2) An overall volume and energy storage capacity estimated respectively at 9560.31 million m³ and 6862.23 GWh (i.e. a power of about 1.9 GW), correspond to all detected sites, where accordingly, the subdivision of Laâyoune-Sakia El Hamra takes a maximum share of 18.59 % and that of Beni Mellal-Khenifra of 27.12 %. (3) As many as 586 high-weighted opportunities have been evaluated nationwide, where the Beni Mellal-Khenifra region accounts for a share of about 20.82 %. These findings highlight the significant opportunities that Morocco enjoys at the national and regional levels. Fig. 11 illustrates these different results. Indeed, the country possesses remarkable solar resources in terms of SPVs [66], whether ground-mounted or floating, as well as in terms of wind energy, either onshore or offshore. As a matter of fact, the offshore potential is much more considerable, estimated at 250 GW, or 10 times more than on land [67]. Setting up a seawater-PHES plant would therefore increase the flexibility of an offshore wind farm and thereby provide a reliable supply. Furthermore, according to Luyao Liu et al. [68], the efficiency of a floating SPV plant is between 1.58 and 2.00 % higher than that of a conventional one under similar ambient conditions. Hence, by taking advantage of the considerable opportunities offered by the chosen locations, the floating SPV plant will increase the system efficiency through the cooling effect, irrespective of the fact that it minimizes the losses associated with reservoir evaporation through shading. These solutions will significantly contribute to meeting the country's growing energy demand by mitigating the intermittency and variability of SPV and wind power. Indeed, according to the projections of the Ministry of Energy Transition and Sustainable Development (METSD), the demand for electricity will reach 95 TWh in 2030 [59]. With PHES systems, about 18 % of the 2020 demand is met, given that the electrical energy consumption has reached 38.7 TWh [69]. As per the Moroccan government's strategic plan for the development of

renewable energies, the share of variable renewables is expected to reach 15 % by 2030 [70]. Accordingly, PHES facilities will support a wide penetration of wind and SPV energy while ensuring a reliable supply and an adjustment between the latter and demand. Moreover, it is worth noting that the electricity provided by the development of these schemes might be used to produce green hydrogen and ammonia, as the country benefits from several phosphate mines (Benguerir, Safi, Khouribga, Youssoufia and Phosboucra).

The foregoing potential configurations highlight the in-situ bulk storage opportunities offered by PHES systems, particularly for the different configurations of SPVs (onshore and floating) and wind (onshore and offshore). However, the scope of the present investigation relies primarily on the evaluation of the potential of PHES systems with respect to the two topologies studied: T2 and T4 within the Moroccan context.

In a nutshell, the developed approach, which is applicable within any case study by adopting the required data, examined two topologies, T2 and T4, which have already demonstrated that Morocco possesses suitable topography for the development of PHES plants. In fact, there are other settings (Table 1) that could also be suitable for the development of this technology, and thereby increase its total potential at the country level. Future work will address the remaining topologies.

5. Conclusions and limitations

The aim of the present study is to establish a methodology to identify potential sites for the development of PHES plants, according to two topologies: T2 and T4. To this end, the physical potential has been evaluated first by taking into account the various specifications as well as the restrictions specified by the model, which incorporates both the diverse topographical requirements, land use and environmental limitations. Afterwards, a multi-criteria assessment process was carried out through the AHP method in order to select the most appropriate sites, taking into account several factors related to the geography, accessibility, risk and climate. The study filled the gap in the decision criteria with two novel factors, namely evapotranspiration and sea surface salinity, neither of which, to our knowledge, have been used in similar studies. This approach has been applied as part of the Moroccan case study, while noting that the model is suitable for application to any territory. In this regard, the existing water bodies have served as one reservoir, as 88 filtered dams and lakes present the first one, where the second candidate reservoirs have been located in a buffer zone of 16 km,

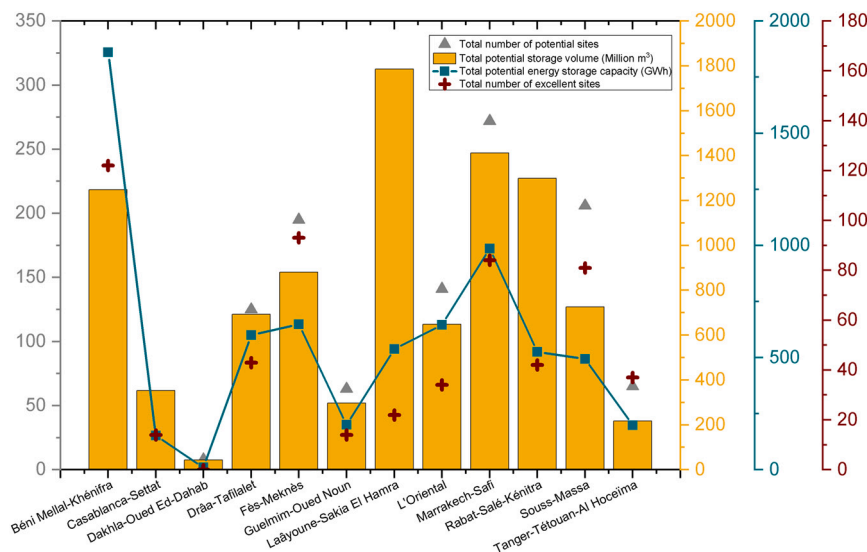


Fig. 11. Total number of potential sites, volume, energy storage capacity and number of excellence sites of PHES plants under T2 and T4 for the 12 regions of Morocco.

for T2. Whereas T4 uses a coastline of >3500 km on two maritime sides: Atlantic: 2934 km, Mediterranean: 512 km to situate the upper reservoir with the same buffering area. The results revealed 936 and 670 potential sites respectively for T2 and T4 with a minimum area of 70,000 m² as stated in the model. These locations offer a total energy storage capacity of approximately 6862.23 GWh per full cycle corresponding to a total estimated volume of around 9602 million m³. Subsequently, a combination analysis was performed involving the various GIS tools with the resulting AHP method weights. The ArcGIS 10.8 software package was used through its variety of tools present in ArcToolboxes as well as with the ModelBuilder feature. The analysis conclusions assigned weights of 59.10 %, 21.79 %, 12.05 %, and 7.06 % to the 4 major criteria: geography, accessibility, risk, and climate accordingly. Where 586 sites are deemed as highly suitable representing 36.50 % of the overall locations. These locations are cost-effective as they have advantageous conditions, leading to higher plant yields while reducing the required investments.

Based on this study, the most promising sites for the development of both conventional and seawater PHES have been evaluated, providing a valuable input for policy makers and stakeholders to select the most cost-effective sites. Nevertheless, and in view of the limitations, it is appropriate to state that a feasibility study considering regulatory policy related to local aspects must be conducted, in order to validate the location before proceeding with deployment. A further limitation is that the resolution of the DEM data used is 3 arcsec (90 m), whereas other available data are more interesting, owing to the significant computation times involved by the computer system used.

With the various solutions provided, it is possible to increase the share of variable renewable energies in the national mix, in line with Moroccan policy. And thus, contribute to the reduction of CO₂ emissions and the achievement of the 2050 decarbonization objectives. In addition, these investments will create additional employment opportunities on a national scale.

CRedit authorship contribution statement

Fatima-zahra Ouchani: Conceptualization, Methodology, Visualization, Data curation, Software, Writing – original draft. **Ouafae Jbahi:** Conceptualization, Visualization, Data curation, Software, Writing – original draft. **Ahmed Alami Merrouni:** Conceptualization, Formal analysis, Software, Visualization, Validation, Writing – original draft. **Abdellatif Ghennioui:** Supervision. **Mohammed Maaroufi:** Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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